Week 8: Database Relationship Analysis Using Joins

Scenario:

An e-commerce company maintains customer, product, order, and payment details in separate tables. The company uses SQL joins to combine related information and analyze customer purchases, order details, and payment records.

Implement INNER JOIN

Concept:

INNER JOIN retrieves only the records that have matching values in both tables.

Tasks: 1. Display customer information along with their orders

```
SQL> SELECT C.*, O.* FROM CUSTOMER C INNER JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                              PHONE      ORDER_ID
-------------------------------------------------- ---------- ----------
ORDER_DAT TOTAL_AMOUNT ORDER_STAT CUSTOMER_ID
--------- ------------ ---------- -----------
       1001 Arun
arun@gmail.com                                     9876543210       1001
27-SEP-26        67500 Pending           1001

       1002 Priya
priya@gmail.com                                    9876543211       1002
27-SEP-26        25000 Shipped           1002

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                              PHONE      ORDER_ID
-------------------------------------------------- ---------- ----------
ORDER_DAT TOTAL_AMOUNT ORDER_STAT CUSTOMER_ID
--------- ------------ ---------- -----------


       1003 Kumar
kumar@gmail.com                                    9876543212       1003
27-SEP-26         1500 Delivered         1003

       1004 Divya
divya@gmail.com                                    9876543213       1004

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                              PHONE      ORDER_ID
-------------------------------------------------- ---------- ----------
ORDER_DAT TOTAL_AMOUNT ORDER_STAT CUSTOMER_ID
--------- ------------ ---------- -----------
27-SEP-26         4200 Pending           1004

       1005 Rahul
rahul@gmail.com                                    9876543214       1005
27-SEP-26         8000 Delivered         1005
```

2.DISPLAY ORDER DETAILS WITH PAYMENT INFORMATION.

```
SQL> SELECT O.Order_ID, O.Order_Date, O.Total_Amount, O.Order_Status, P.Payment_ID, P.Payment_Date, P.Payment_Mode, P.Payment_Status, P.Transaction_Amount FROM Orders O INNER JOI
N Payment P ON O.Order_ID = P.Order_ID;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STAT PAYMENT_ID PAYMENT_D
---------- --------- ------------ ---------- ---------- ---------
PAYMENT_MODE         PAYMENT_ST TRANSACTION_AMOUNT
-------------------- ---------- ------------------
      1001 27-SEP-26        67500 Pending            1 27-SEP-26
UPI                  Success                67500

      1001 27-SEP-26        67500 Pending            6 29-SEP-26
Card                 Failed                  1000

      1002 27-SEP-26        25000 Shipped            2 27-SEP-26
Card                 Success                25000


  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STAT PAYMENT_ID PAYMENT_D
---------- --------- ------------ ---------- ---------- ---------
PAYMENT_MODE         PAYMENT_ST TRANSACTION_AMOUNT
-------------------- ---------- ------------------
      1002 27-SEP-26        25000 Shipped            7 30-SEP-26
UPI                  Pending                 5000

      1003 27-SEP-26         1500 Delivered          3 28-SEP-26
Cash                 Success                 1500

      1003 27-SEP-26         1500 Delivered          8 30-SEP-26
Net Banking          Failed                  2000


  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STAT PAYMENT_ID PAYMENT_D
---------- --------- ------------ ---------- ---------- ---------
PAYMENT_MODE         PAYMENT_ST TRANSACTION_AMOUNT
-------------------- ---------- ------------------
      1004 27-SEP-26         4200 Pending            4 28-SEP-26
Net Banking          Pending                 4200

      1005 27-SEP-26         8000 Delivered          5 29-SEP-26
UPI                  Success                 8000
```

3.RETRIEVE PRODUCT PURCHASED BY CUSTOOMERS

```
SQL> SELECT C.Customer_ID, C.Customer_Name, O.Order_ID, P.Product_ID, P.Product_Name, P.Price FROM Customer C INNER JOIN Orders O ON C.Customer_ID = O.Customer_ID INNER JOIN Orde
r_Details OD ON O.Order_ID = OD.Order_ID INNER JOIN Product P ON OD.Product_ID = P.Product_ID;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ---------
PRODUCT_ID PRODUCT_NAME              PRICE
---------- -------------------- ----------
       1001 Arun                                                   1001
P1001      Laptops                   67500

       1002 Priya                                                  1002
P1002      Mobile                    25000

       1003 Kumar                                                  1003
P1003      TShirt                      800


CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ---------
PRODUCT_ID PRODUCT_NAME              PRICE
---------- -------------------- ----------
       1004 Divya                                                  1004
P1004      Shoes                     1500
```

IMPLEMENT LEFT JOIN

TASKS:

1.DISPLAY ALL CUSTOMERS INCLUDING CUSTOMERS WHO HAVE NOT PLACED ORDERS.

```
SQL> SELECT C.Customer_ID, C.Customer_Name, C.Email, C.Phone, O.Order_ID, O.Order_Date, O.Total_Amount, O.Order_Status FROM Customer C LEFT JOIN Orders O ON C.Customer_ID = O.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                              PHONE      ORDER_ID
-------------------------------------------------- ---------- ----------
ORDER_DAT TOTAL_AMOUNT ORDER_STAT
--------- ------------ ----------
       1001 Arun
arun@gmail.com                                     9876543210       1001
27-SEP-26        67500 Pending

       1002 Priya
priya@gmail.com                                    9876543211       1002
27-SEP-26        25000 Shipped

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                              PHONE      ORDER_ID
-------------------------------------------------- ---------- ----------
ORDER_DAT TOTAL_AMOUNT ORDER_STAT
--------- ------------ ----------

       1003 Kumar
kumar@gmail.com                                    9876543212       1003
27-SEP-26         1500 Delivered

       1004 Divya
divya@gmail.com                                    9876543213       1004

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                              PHONE      ORDER_ID
-------------------------------------------------- ---------- ----------
ORDER_DAT TOTAL_AMOUNT ORDER_STAT
--------- ------------ ----------
27-SEP-26         4200 Pending

       1005 Rahul
rahul@gmail.com                                    9876543214       1005
27-SEP-26         8000 Delivered
```

2.FIND CUSTOMERS WITHOUT ANY PURCHASES.

```
SQL> SELECT C.Customer_ID, C.Customer_Name, C.Email, C.Phone FROM Customer C LEFT JOIN Orders O
ON C.Customer_ID = O.Customer_ID WHERE O.Order_ID IS NULL ORDER BY C.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                                    PHONE
-------------------------------------------------- ----------
       1006 Rohan
rohan@gmail.com                                    9876543215
```

3.DISPLAY ALL PRODUCTS INCLUDING PRODUCTS WITH NO SALES.

```
SQL> SELECT P.Product_ID, P.Product_Name, P.Price FROM Product P LEFT JOIN Order_Details OD ON P
.Product_ID = OD.Product_ID WHERE OD.Product_ID IS NULL ORDER BY P.Product_ID;

PRODUCT_ID PRODUCT_NAME              PRICE
---------- -------------------- ----------
P1006      Mouse                       500
```

IMPLEMENT RIGHT JOIN

TASK:

1.DISPLAY ALL ORDERS WITH CUSTOMERS INFORMATION

```
SQL> SELECT O.Order_ID, O.Order_Date, O.Total_Amount, O.Order_Status, C.Customer_ID, C.Customer_Name, C.Email FROM Customer C RIGHT JOIN Orders O ON C.Customer_ID = O.Customer_ID
 ORDER BY O.Order_Date;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STAT CUSTOMER_ID
---------- --------- ------------ ---------- -----------
CUSTOMER_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------
      1001 27-SEP-26        67500 Pending           1001
Arun
arun@gmail.com

      1002 27-SEP-26        25000 Shipped           1002
Priya
priya@gmail.com

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STAT CUSTOMER_ID
---------- --------- ------------ ---------- -----------
CUSTOMER_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------

      1003 27-SEP-26         1500 Delivered         1003
Kumar
kumar@gmail.com

      1004 27-SEP-26         4200 Pending           1004
Divya

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STAT CUSTOMER_ID
---------- --------- ------------ ---------- -----------
CUSTOMER_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------
divya@gmail.com

      1005 27-SEP-26         8000 Delivered         1005
Rahul
rahul@gmail.com
```

2.FIND ORDERS WHERE CUSTOMERS DETAILS ARE MISSING.

```
SQL> SELECT O.Order_ID, O.Order_Date, O.Total_Amount, O.Order_Status, O.Customer_ID FROM Customer C RIGHT JOIN Or
ders O ON C.Customer_ID = O.Customer_ID WHERE C.Customer_ID IS NULL;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STAT CUSTOMER_ID
---------- --------- ------------ ---------- -----------
      1006 25-OCT-26         1500 Pending            9999
```

3.DISPLAY ALL PAYMENT RECORDS WITH ORDER DETAILS.

```
SQL> SELECT P.Payment_ID, P.Order_ID, P.Payment_Date, P.Payment_Mode, P.Payment_Status, P.Transaction_Amount, O.Order_Date, O.Total_Amount, O.Order_Status FROM Payment P INNER JOIN Orders O ON P.Order_ID = O.
Order_ID ORDER BY P.Payment_ID;

PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_ST
---------- ---------- --------- -------------------- ----------
TRANSACTION_AMOUNT ORDER_DAT TOTAL_AMOUNT ORDER_STAT
------------------ --------- ------------ ----------
         1       1001 27-SEP-26 UPI                  Success
             67500 27-SEP-26        67500 Pending

         2       1002 27-SEP-26 Card                 Success
             25000 27-SEP-26        25000 Shipped

         3       1003 28-SEP-26 Cash                 Success
              1500 27-SEP-26         1500 Delivered


PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_ST
---------- ---------- --------- -------------------- ----------
TRANSACTION_AMOUNT ORDER_DAT TOTAL_AMOUNT ORDER_STAT
------------------ --------- ------------ ----------
         4       1004 28-SEP-26 Net Banking          Pending
              4200 27-SEP-26         4200 Pending

         5       1005 29-SEP-26 UPI                  Success
              8000 27-SEP-26         8000 Delivered

         6       1001 29-SEP-26 Card                 Failed
              1000 27-SEP-26        67500 Pending


PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_ST
---------- ---------- --------- -------------------- ----------
TRANSACTION_AMOUNT ORDER_DAT TOTAL_AMOUNT ORDER_STAT
------------------ --------- ------------ ----------
         7       1002 30-SEP-26 UPI                  Pending
              5000 27-SEP-26        25000 Shipped

         8       1003 30-SEP-26 Net Banking          Failed
              2000 27-SEP-26         1500 Delivered


8 rows selected.
```

5.RETRIEVE COMPLETE ORDERS DETAILS

TASK:

GENERAATE A COMPLETE ORDER REPORT BY COMBINING:

- CUSTOMER NAME
- PRODUCT NAME
- QUANTITY
- ORDER DATE
- TOTAL AMOUNT
- PAYMENT STATUS

```
SQL> SELECT C.Customer_Name, P.Product_Name, OD.Quantity, O.Order_Date, O.Total_Amount, PY.Payment_Status FROM Customer C JOIN Orders O ON C.Customer_ID = O.Customer_ID JOIN Order_Details OD ON O.Order_ID = O
D.Order_ID JOIN Product P ON OD.Product_ID = P.Product_ID LEFT JOIN Payment PY ON O.Order_ID = PY.Order_ID ORDER BY O.Order_ID;

CUSTOMER_NAME                                      PRODUCT_NAME
-------------------------------------------------- --------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------
Arun                                               Laptops
         1 27-SEP-26        67500 Success

Arun                                               Laptops
         1 27-SEP-26        67500 Failed

Priya                                              Mobile
         1 27-SEP-26        25000 Success


CUSTOMER_NAME                                      PRODUCT_NAME
-------------------------------------------------- --------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------
Priya                                              Mobile
         1 27-SEP-26        25000 Pending

Kumar                                              TShirt
         2 27-SEP-26         1500 Success

Kumar                                              TShirt
         2 27-SEP-26         1500 Failed


CUSTOMER_NAME                                      PRODUCT_NAME
-------------------------------------------------- --------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_ST
---------- --------- ------------ ----------
Divya                                              Shoes
         2 27-SEP-26         4200 Pending

Rahul                                              Keyboard
         1 27-SEP-26         8000 Success

8 rows selected.
```

6.DISPLAY CUSTOMER PURCHASE HISTORY

TASKS:

1.ALL PRODUCTS PURCHASED BY A CUSTOMER.

```
SQL> SELECT C.Customer_ID, C.Customer_Name, P.Product_ID, P.Product_Name, OD.Quantity, OD.Price FROM Customer C JOIN Orders O ON C.Custome
r_ID = O.Customer_ID JOIN Order_Details OD ON O.Order_ID = OD.Order_ID JOIN Product P ON OD.Product_ID = P.Product_ID ORDER BY C.Customer_
ID;

CUSTOMER_ID CUSTOMER_NAME                                      PRODUCT_ID
----------- -------------------------------------------------- ----------
PRODUCT_NAME           QUANTITY      PRICE
-------------------- ---------- ----------
       1001 Arun                                              P1001
Laptops                     1      67500

       1002 Priya                                             P1002
Mobile                      1      25000

       1003 Kumar                                             P1003
TShirt                      2        750


CUSTOMER_ID CUSTOMER_NAME                                      PRODUCT_ID
----------- -------------------------------------------------- ----------
PRODUCT_NAME           QUANTITY      PRICE
-------------------- ---------- ----------
       1004 Divya                                             P1004
Shoes                       2       2100

       1005 Rahul                                             P1005
Keyboard                    1       8000
```

2.TOTAL AMOUNT SPENT BY EACH CUSTOMERS.

```
SQL> SELECT C.Customer_ID, C.Customer_Name, SUM(O.Total_Amount) AS Total_Spent FROM Customer C JOIN Orders O ON C.Customer_ID = O.Customer
_ID GROUP BY C.Customer_ID, C.Customer_Name ORDER BY C.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                      TOTAL_SPENT
----------- -------------------------------------------------- -----------
       1001 Arun                                                     67500
       1002 Priya                                                    25000
       1003 Kumar                                                     1500
       1004 Divya                                                     4200
       1005 Rahul                                                     8000
```

3.NUMBER OF ORDERS PLACED BY EACH CUSTOMER

```
SQL> SELECT C.Customer_ID, C.Customer_Name, COUNT(O.Order_ID) AS Total_Orders FROM Customer C LEFT JOIN Orders O ON C.Customer_ID = O.Cust
omer_ID GROUP BY C.Customer_ID, C.Customer_Name ORDER BY C.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                      TOTAL_ORDERS
----------- -------------------------------------------------- ------------
       1001 Arun                                                         1
       1002 Priya                                                        1
       1003 Kumar                                                        1
       1004 Divya                                                        1
       1005 Rahul                                                        1
       1006 Rohan                                                        0

6 rows selected.
```

4.LATEST PURCHASE DETAILS OF CUSTOMERS.

```
SQL> SELECT C.Customer_ID, C.Customer_Name, O.Order_ID, O.Order_Date AS Latest_Purchase_Date, O.Total_Amount, O.Order_Status FROM Customer
 C JOIN Orders O ON C.Customer_ID = O.Customer_ID WHERE O.Order_Date = (SELECT MAX(O2.Order_Date) FROM Orders O2 WHERE O2.Customer_ID = C.
Customer_ID) ORDER BY C.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                       ORDER_ID
----------- -------------------------------------------------- ----------
LATEST_PU TOTAL_AMOUNT ORDER_STAT
--------- ------------ ----------
       1001 Arun                                                     1001
27-SEP-26        67500 Pending

       1002 Priya                                                    1002
27-SEP-26        25000 Shipped

       1003 Kumar                                                    1003
27-SEP-26         1500 Delivered


CUSTOMER_ID CUSTOMER_NAME                                       ORDER_ID
----------- -------------------------------------------------- ----------
LATEST_PU TOTAL_AMOUNT ORDER_STAT
--------- ------------ ----------
       1004 Divya                                                    1004
27-SEP-26         4200 Pending

       1005 Rahul                                                    1005
27-SEP-26         8000 Delivered
```

7.GENERATE MULTI-TABLE BUSINESS REPORTS

REPORT 1:CUSTOMER ORDER REPORT

DISPLAY:

- CUSTOMER NAME
- ORDER ID
- ORDER DATE
- ORDER STATUS

```
SQL> SELECT C.Customer_Name, O.Order_ID, O.Order_Date, O.Order_Status FROM Customer C INNER JOIN Orders O ON C.Customer_ID = O.Customer_ID
 ORDER BY O.Order_Date;

CUSTOMER_NAME                                       ORDER_ID ORDER_DAT
--------------------------------------------------- ---------- ---------
ORDER_STAT
----------
Arun                                                      1001 27-SEP-26
Pending

Priya                                                     1002 27-SEP-26
Shipped

Kumar                                                     1003 27-SEP-26
Delivered


CUSTOMER_NAME                                       ORDER_ID ORDER_DAT
--------------------------------------------------- ---------- ---------
ORDER_STAT
----------
Divya                                                     1004 27-SEP-26
Pending

Rahul                                                     1005 27-SEP-26
Delivered
```

REPORT 2:SALES REPORT

DISPLAY:

- PROUDUCT NAME
- QUANTITY SOLD
- TOTAL REVENUE

```
SQL> SELECT P.Product_Name, SUM(OD.Quantity) AS Quantity_Sold, SUM(OD.Quantity * OD.Price) AS Total_Revenue FROM Product P INNER JOIN Orde
r_Details OD ON P.Product_ID = OD.Product_ID GROUP BY P.Product_ID, P.Product_Name ORDER BY Total_Revenue DESC;

PRODUCT_NAME         QUANTITY_SOLD TOTAL_REVENUE
-------------------- ------------- -------------
Laptops                          1         67500
Mobile                           1         25000
Keyboard                         1          8000
Shoes                            2          4200
TShirt                           2          1500
```

REPORT 3: PAYMENT ANALYSIS REPORT

DISPLAY:

- PAYMENT MODE
- NUMBER OF TRANSACTION
- SUCCESSFUL PAYMENTS

```
SQL> SELECT Payment_Mode, COUNT(*) AS Number_Of_Transactions, SUM(CASE WHEN Payment_Status = 'Success' THEN 1 ELSE 0 END) AS Successful_Pa
yments FROM Payment GROUP BY Payment_Mode ORDER BY Payment_Mode;

PAYMENT_MODE         NUMBER_OF_TRANSACTIONS SUCCESSFUL_PAYMENTS
-------------------- ---------------------- -------------------
Card                                      2                   1
Cash                                      1                   1
Net Banking                               2                   0
UPI                                       3                   2
```

REPORT 4: CUSTOMER PURCHASE ANALYSIS

FIND:

TOP PURCHASING CUSTOMERS

```
SQL> SELECT C.Customer_ID, C.Customer_Name, SUM(O.Total_Amount) AS Total_Purchase FROM Customer C JOIN Orders O ON C.Customer_ID = O.Custo
mer_ID GROUP BY C.Customer_ID, C.Customer_Name ORDER BY Total_Purchase DESC;

CUSTOMER_ID CUSTOMER_NAME                                          TOTAL_PURCHASE
----------- -------------------------------------------------- --------------
       1001 Arun                                                        67500
       1002 Priya                                                       25000
       1005 Rahul                                                        8000
       1004 Divya                                                        4200
       1003 Kumar                                                        1500
```

CUSTOMERS WITH MAXIMUM ORDERS

```
SQL> SELECT C.Customer_ID, C.Customer_Name, COUNT(O.Order_ID) AS Total_Orders FROM Customer C JOIN Orders O ON C.Customer_ID = O.Customer_
ID GROUP BY C.Customer_ID, C.Customer_Name HAVING COUNT(O.Order_ID) = (SELECT MAX(Order_Count) FROM (SELECT COUNT(*) AS Order_Count FROM O
rders GROUP BY Customer_ID)) ORDER BY C.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                          TOTAL_ORDERS
----------- -------------------------------------------------- ------------
       1001 Arun                                                           1
       1002 Priya                                                          1
       1003 Kumar                                                          1
       1004 Divya                                                          1
       1005 Rahul                                                          1
```

CUSTOMERS WITH HIGHEST SPENDING

```
SQL> SELECT C.Customer_ID, C.Customer_Name, SUM(O.Total_Amount) AS Total_Spending FROM Customer C JOIN Orders O ON C.Customer_ID = O.Custo
mer_ID GROUP BY C.Customer_ID, C.Customer_Name HAVING SUM(O.Total_Amount) = (SELECT MAX(Total_Spending) FROM (SELECT SUM(Total_Amount) AS
Total_Spending FROM Orders GROUP BY Customer_ID));

CUSTOMER_ID CUSTOMER_NAME                                      TOTAL_SPENDING
----------- -------------------------------------------------- --------------
       1001 Arun                                                       67500
```